
Energy Efficient Patent Classification on Text-attributed Graph

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1 Problem Statement

Patent classification is one of the major tasks in patent life-cycle. It is a method used to systematically categorize U.S. patent documents into distinct technology groups based on subject matter ¹. A text-attributed graph (TAG) consists of nodes with textual attributes or descriptions and edges represent relationships between nodes. Node classification on TAG involves classifying the node based on both textual and structural representations. In the context of patent data, each patent—defined by its title, abstract, or claims—is treated as a node and citation links between patents form the edges. This dual-modality is crucial for patent analysis because the text provides detailed semantic information about the invention, and the citation network offers insights into how patents build upon each other. However, implementing deep learning models, such as Graph Neural Networks (GNNs) combined with Language models (LMs), on such vast datasets poses significant challenges. Patent datasets often consists of large numbers of nodes and edges which will need high computational costs and energy consumption when training and generating embeddings. Thus, a key challenge is to develop energy-efficient classification methods that effectively leverage both the textual and graph structures of patent data and reduce model complexity while preserving or even enhancing classification accuracy [8].

2 Brief Survey and Uniqueness

Survey. Text-attributed graph classification has seen significant progress by integrating deep learning techniques with graph structures. Early approaches leveraged Graph Neural Networks such as GCNs [5] and GraphSAGE [2] that used static text embeddings to represent node features, while more recent studies have incorporated language models to extract richer semantic representations. Notably, recent work utilizes hybrid models that combine pre-trained language models with GNNs in cascaded or end-to-end architectures to learn from both text and graph modalities [9]. The field has focused on reducing the computational and energy burdens of these models. Techniques such as pruning [6] and quantization [7] lower model complexity and accelerate inference, while knowledge distillation [3] helps in transferring the performance of large models to more compact ones.

Despite advancements in representing TAG across various domains, the patent domain remains unexplored. The availability of patent graph datasets is a significant bottleneck. Existing patent citation datasets often lack full-text information, and text-based patent datasets rarely include structured graph relationships. Although a few approaches have been taken for patent classification [4, 1], none have effectively utilized graph structures. This gap in readily available, comprehensive text-attributed patent datasets has motivated us to build a text-attributed patent graph and apply energy-efficient deep learning models to solve patent classification and related tasks.

Uniqueness. Patent data is a valuable resource for understanding industry trends and guiding research investments. Efficient analysis of patent information not only streamline the review process but also helps innovators in their research and development efforts. Key patent analysis tasks include

¹<https://www.uspto.gov/patents/search/classification-standards-and-development>

classification, retrieval, and novelty detection. This work will introduce a novel text-attributed patent graph dataset and develop an energy-efficient hybrid model compared to the existing approaches that integrates a language model with a graph neural network to solve patent classification and related tasks. Our approach will prioritize efficiency through the combination of existing energy-efficient methods into the model development pipeline and will show a comparison of different techniques. This work aims to achieve robust classification performance with significantly lower energy consumption and address the gap in current research on patent analysis.

3 Preliminary plans & Milestones

Weeks 1-2: Build a patent dataset by collecting and preprocessing available USPTO patent text and citation data. Conduct an in-depth literature review on text-attributed graph learning and energy-efficient techniques. Define the target classification task.

Weeks 3-4: Develop a structured patent graph dataset and implement a baseline model combining a text encoder with a GNN; prepare and submit the intermediate project report.

Weeks 5-6: Implement the baseline model and progressively integrate energy-efficient combination strategies; evaluate early performance and efficiency metrics.

Weeks 7-8: Refine the model through evaluation and optimization to balance classification accuracy and energy efficiency, and finalize the project report for submission.

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